Notes: Whited and Zhao (JF, 2021)

The Misallocation of Finance

Contents

1 Big picture

- **Question.** Are financial liabilities misallocated across firms? More specifically, do firms get the *right total amount of finance*, and do they get the *right mix of debt and equity*?
- **Core idea.** The paper takes the Hsieh–Klenow misallocation framework, which is usually written on the asset or factor side of the balance sheet, and moves it to the liability side. Debt and equity are treated as the primitive inputs that ultimately finance capital, labor, materials, and other proximate factors.
- **Main modeling contribution.** Debt and equity are combined through a CES aggregator, so the paper can distinguish between two sources of inefficiency: misallocation of *scale* (too much or too little total finance for a given firm) and misallocation of *type* (the wrong debt–equity mix).
- **Main empirical result.** Financial liabilities look relatively well allocated in the United States but not in China. Relative to the U.S. benchmark, reallocating Chinese liabilities would raise real value-added by about 51% to 69%.
- **Most important decomposition.** Roughly 79% to 83% of the Chinese gains come from the *amount* of finance, while only 17% to 21% come from the debt–equity mix itself.
- **Economic intuition.** The biggest problem is not mainly that firms are financed with too much debt instead of equity, or vice versa. It is that productive firms are too small and less productive firms are too large because the gross flow of financial resources is mismatched to firm productivity.
- **Why the paper matters.** It offers a tractable, mostly empirical way to quantify how much a poorly functioning financial system can lower aggregate output, without having to specify every primitive friction in a fully dynamic equilibrium model.
- **Important caveat.** This is a model-based measurement exercise, not a standard causal research design. Because the static framework can be misspecified, the paper treats U.S. allocations as a useful benchmark and reports China relative to U.S. efficiency.

2 Data and measurement (key objects)

2.1 Chinese manufacturing data

- **Source.** Annual survey of industrial firms from the National Bureau of Statistics (NBS) of China.
- **Period.** 1999–2007.
- **Sector focus.** Manufacturing firms only. The mining sector is excluded because it is small and operationally different, and utilities are excluded because they are heavily regulated.
- **Ownership screen.** The baseline sample excludes state-owned and collective corporations. A firm-year is classified as private if state and collective paid-in capital together account for less than 50%.

- **Size screen.** Firms with less than 5 million 1999 CNY in sales are dropped to reduce selection problems from incomplete reporting by very small firms.
- **Additional screens.** The paper drops observations with negative or missing value-added, total liabilities, or shareholder equity.
- **Final sample size.** 1,248,729 firm-year observations.

2.2 Chinese variable construction

- **Debt measure.** D_{si} is total liabilities, not a narrower debt item.
- **Equity measure.** E_{si} is shareholders' equity.
- **Why total liabilities?** First, "debt" is not directly available in the Chinese data. Second, with a CES aggregator, zero debt observations would create problems because marginal benefits blow up at zero.
- **Nominal benefit of finance.** $P_{si}F_{si}$ is nominal value-added, computed as profits plus indirect taxes plus wages plus depreciation.
- **Real value-added for estimating elasticities.** The paper constructs a firm-specific output deflator using current-price and constant-price industrial output, available in the Chinese data for 1999–2003.

2.3 Chinese summary-statistic patterns

- **Leverage and size.** The ratio of liabilities to assets rises with firm size, from about 0.483 in the bottom size bin to 0.569 in the top size bin.
- **Productivity and size.** Smaller firms generate more value-added per unit of assets than larger firms. This is the first descriptive hint that finance may be going to the wrong firms.
- **Time pattern.** Average leverage is fairly stable over time, but average value-added rises noticeably by the end of the sample.

2.4 U.S. manufacturing data

- **Source.** Compustat.
- **Period.** 1999–2007, to match the Chinese sample.
- **Sector focus.** Manufacturing firms with SIC codes 2000–3999.
- **Final sample size.** 14,158 firm-year observations.
- **Debt and equity.** D_{si} is total liabilities (LT) and E_{si} is shareholders' equity, computed as assets minus liabilities (AT–LT).
- **Nominal value-added.** Operating income before depreciation (OIBDP) plus imputed wages.
- **Wage imputation.** Mean wage per employee is computed from the NBER–CES Manufacturing Industry Database at the three-digit SIC level and then multiplied by firm employment.

- **Real value-added for elasticity estimation.** Nominal value-added is deflated by the shipments price deflator from the NBER productivity database.

2.5 U.S. summary-statistic patterns

- **Leverage and size.** As in China, leverage rises with size.
- **Value-added relative to assets.** Unlike China, both small and large U.S. firms display roughly similar value-added-to-assets ratios.
- **Time pattern.** U.S. firms grow over time and become somewhat less leveraged.
- **Interpretation.** The U.S. cross-section looks much more consistent with liabilities being allocated in line with firm productivity.

2.6 Sector definitions and empirical objects

- **Aggregation.** Reallocation gains are computed within three-digit industries and then aggregated.
- **Elasticity estimation.** The debt–equity substitution parameter γ_s is estimated at the two-digit sector level.
- **Main outcome.** The paper's summary statistic for efficiency is the *fractional benefit* $F/\widehat{F}$, that is, actual real benefit divided by efficiently reallocated real benefit.
- **Percent gain from reallocation.** If the fractional benefit is $F/\widehat{F}$, then the associated gain is $(\widehat{F}/F) - 1$.

3 Models and empirical specifications (all equations + interpretation)

3.1 Aggregate benefit across sectors

The economy-wide real benefit of finance is a Cobb–Douglas aggregate across sectors:

$$F = \prod_{s=1}^{S} F_s^{\theta_s}, \qquad \sum_{s=1}^{S} \theta_s = 1. \tag{1}$$

- **Interpretation.** The economy values finance-generated output from all sectors, but with diminishing marginal benefit to making any one sector arbitrarily large.
- **Role of θ_s.** These are sector expenditure shares in the aggregate economy.

3.2 Sector-level aggregation across firms

Within sector s, the real benefit is a CES aggregate across differentiated firms:

$$F_s = \left(\sum_{i=1}^{I} F_{si}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}. \tag{2}$$

- **Interpretation.** Firms are monopolistically competitive. The elasticity of substitution σ controls how strongly the market reallocates demand toward more productive firms when their relative prices fall.
- **Economic meaning of σ.** A larger σ means more substitutable firm outputs and therefore potentially larger gains from reallocating finance toward better firms.

3.3 Firm-level technology: debt and equity as primitive inputs

At the firm level, debt and equity are combined through a CES technology:

$$F_{si} = A_{si} \left[\alpha_s D_{si}^{\frac{\gamma_s-1}{\gamma_s}} + (1-\alpha_s) E_{si}^{\frac{\gamma_s-1}{\gamma_s}} \right]^{\frac{\gamma_s}{\gamma_s-1}}. \tag{3}$$

- **Interpretation.** A_{si} is firm-level total factor benefit (TFB), the liability-side analogue of TFP. Debt and equity finance combine to generate real value-added.
- **Key parameters.** α_s is the importance weight on debt; γ_s is the elasticity of substitution between debt and equity.
- **Why CES matters.** This is the paper's key extension relative to Hsieh–Klenow. If γ_s is finite, debt and equity are imperfect substitutes and the mix matters. If $\gamma_s \to \infty$, they become perfect substitutes, reproducing a Modigliani–Miller benchmark.
- **Modeling interpretation.** The paper treats capital, labor, materials, and energy as unmodeled intermediate inputs financed by liabilities.

3.4 Nominal profit and financing wedges

The firm chooses finance to maximize nominal profit (or economic value-added):

$$\pi_{si} = P_{si} F_{si} - \left[(1+\tau_{si}^D) r_{si} D_{si} + (1+\tau_{si}^E) \lambda_{si} E_{si} \right]. \tag{4}$$

- **Interpretation.** r_{si} and λ_{si} are the underlying costs of debt and equity, while τ_{si}^D and τ_{si}^E are reduced-form wedges that capture distortions.
- **Economic meaning of wedges.** Positive wedges are extra costs generated by frictions such as asymmetric information, agency problems, covenant constraints, collateral scarcity, or credit rationing. Negative wedges capture subsidies or favored access.
- **Important feature.** The paper does not commit to one primitive friction. The wedges summarize all distortions relevant for the observed financing allocation.

3.5 Optimal debt–equity mix inside the firm

Let total financing cost be

$$C_{si}(D_{si}, E_{si}) = (1 + \tau_{si}^D) r_{si} D_{si} + (1 + \tau_{si}^E) \lambda_{si} E_{si}.$$

The optimal debt–equity ratio solves

$$Z_{si} \equiv \frac{D_{si}}{E_{si}} = \left[\frac{\alpha_s}{1 - \alpha_s} \frac{\partial C_{si} / \partial E_{si}}{\partial C_{si} / \partial D_{si}} \right]^{\gamma_s}. \tag{5}$$

- **Interpretation.** The firm chooses debt and equity so that their marginal net benefits are equalized.
- **Comparative static.** If debt becomes more expensive at the margin relative to equity, the optimal debt–equity ratio falls.
- **Key empirical implication.** Large dispersion in leverage ratios within a sector is evidence of distorted marginal financing costs relative to the benchmark model.

3.6 Marginal cost of one unit of the real benefit and pricing

Given the optimal ratio Z_{si}, the minimized marginal financing cost of producing one unit of F_{si} is

$$\begin{aligned} c_{si} = \frac{1}{A_{si}} \Bigg[& (1 + \tau_{si}^D) r_{si} \left(\alpha_s + (1 - \alpha_s) Z_{si}^{-\frac{\gamma_s - 1}{\gamma_s}} \right)^{-\frac{\gamma_s}{\gamma_s - 1}} \\ & + (1 + \tau_{si}^E) \lambda_{si} \left(\alpha_s Z_{si}^{\frac{\gamma_s - 1}{\gamma_s}} + (1 - \alpha_s) \right)^{-\frac{\gamma_s}{\gamma_s - 1}} \Bigg]. \end{aligned} \tag{6}$$

The firm then charges a constant markup over marginal cost:

$$P_{si} = \frac{\sigma}{\sigma - 1} c_{si}. \tag{7}$$

- **Interpretation.** Price is a standard Dixit–Stiglitz markup over minimized marginal cost.
- **Economic point.** More productive firms (high A_{si}) or firms facing smaller wedges can support lower marginal cost and thus lower price at a given markup.

3.7 Sector and aggregate price indices

The sector price index is

$$P_s = \left(\sum_{i=1}^{I} P_{si}^{-(\sigma - 1)} \right)^{-\frac{1}{\sigma - 1}}, \tag{8}$$

and the aggregate price level is

$$P = \prod_{s=1}^{S} \left(\frac{P_s}{\theta_s} \right)^{\theta_s}. \tag{9}$$

Sector expenditure shares satisfy

$$P_s F_s = \theta_s P F. \tag{10}$$

- **Interpretation.** These equations pin down how firm-level distortions aggregate up to sectoral and then economy-wide efficiency losses.

3.8 Recovering firm productivity from observables

Because firm-level real benefit F_{si} is unobserved, the paper backs out A_{si} from nominal value-added and observed liabilities:

$$A_{si} = \eta_s \frac{(P_{si} F_{si})^{\frac{\sigma}{\sigma-1}}}{\left[\alpha_s D_{si}^{\frac{\gamma_s-1}{\gamma_s}} + (1-\alpha_s) E_{si}^{\frac{\gamma_s-1}{\gamma_s}} \right]^{\frac{\gamma_s}{\gamma_s-1}}}, \qquad \eta_s = \frac{1}{P_s (P_s F_s)^{\frac{1}{\sigma-1}}}. \tag{11}$$

- **Interpretation.** TFB is inferred from observed value-added and financing choices, using the model's structure to translate nominal outcomes into real productivity.
- **Normalization.** Since η_s is common to all firms in a sector, it cancels out in the reallocation exercise and is normalized to one.

3.9 Efficient reallocation: the planner's allocation

Under the efficient allocation, all firms in sector s share the same optimal debt–equity ratio:

$$\frac{\widehat{D}_{si}}{\widehat{E}_{si}} = \frac{D_s}{E_s}. \tag{12}$$

The planner reallocates the total amounts of debt and equity in proportion to firm productivity:

$$\widehat{D}_{si} = \frac{A_{si}^{\sigma-1}}{\sum_j A_{sj}^{\sigma-1}} D_s, \qquad \widehat{E}_{si} = \frac{A_{si}^{\sigma-1}}{\sum_j A_{sj}^{\sigma-1}} E_s. \tag{13}$$

Efficient firm, sector, and aggregate real benefits are then

$$\widehat{F}_{si} = A_{si} \left[\alpha_s \widehat{D}_{si}^{\frac{\gamma_s-1}{\gamma_s}} + (1-\alpha_s) \widehat{E}_{si}^{\frac{\gamma_s-1}{\gamma_s}} \right]^{\frac{\gamma_s}{\gamma_s-1}}, \tag{14}$$

$$\widehat{F}_s = \left(\sum_i \widehat{F}_{si}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \qquad \widehat{F} = \prod_s \widehat{F}_s^{\theta_s}. \tag{15}$$

- **Interpretation.** Efficient firms receive more debt and more equity when they have higher TFB.
- **Crucial result.** The planner equalizes the debt–equity ratio within a sector, but also scales total finance across firms in proportion to productivity.
- **Efficiency metric.** Reallocation gains are summarized by the fractional benefit $F/\widehat{F}$.

3.10 MM benchmark and why it matters

In the Modigliani–Miller limit, $\sigma \to \infty$, $\gamma_s \to \infty$, and $\tau_{si}^D = \tau_{si}^E = 0$. Then nominal profit collapses to zero:

$$\pi_{si} = P_{si}F_{si} - (r_{si}D_{si} + \lambda_{si}E_{si}) = 0. \tag{16}$$

The nominal benefit of finance depends only on total liabilities:

$$\frac{P_{si}F_{si}}{u_{si}} = D_{si} + E_{si}, \tag{17}$$

where u_{si} is the average cost of total finance,

$$u_{si} = r_{si}\frac{D_{si}}{D_{si} + E_{si}} + \lambda_{si}\frac{E_{si}}{D_{si} + E_{si}}. \tag{18}$$

This yields the standard MM Proposition II relation:

$$\lambda_{si} = u_{si} + \frac{D_{si}}{E_{si}}(u_{si} - r_{si}). \tag{19}$$

- **Interpretation.** In an MM world, the debt–equity mix itself is irrelevant. Only total finance matters.
- **Why important here.** This limit lets the authors split total misallocation into misallocation of *scale* and misallocation of *type*.

3.11 The roles of σ and γ_s

- **Elasticity across firms (σ).** Reallocation gains increase in σ. If firm outputs are close substitutes, productive firms can expand a lot when they receive more finance without facing a very steep demand penalty.
- **Elasticity between debt and equity (γ_s).** Reallocation gains from changing the debt–equity mix decrease in γ_s. When debt and equity are close substitutes, even sizable leverage differences have small efficiency consequences.
- **Scale versus type.** Even if $\gamma_s = \infty$, scale misallocation can remain if total finance is not allocated in proportion to productivity. This is why the paper connects its "scale" component to the usual real misallocation in Hsieh–Klenow.
- **Interpretation of γ_s.** The paper views γ_s as partly reflecting sector-wide technological and financial structure, while the wedges τ_{si}^D and τ_{si}^E capture firm-specific distortions.

3.12 Estimating the substitution elasticity and calibrating demand elasticity

- **Estimating γ_s.** The paper estimates γ_s separately by two-digit sector using a Kmenta-style approximation for CES production functions applied to real value-added, debt, and equity.
- **Why estimate γ_s separately?** This avoids using the same variation both to pin down sector-wide substitutability and to infer firm-specific wedges.
- **Baseline identifying assumption.** In the fixed-effects specification, firm productivity A_{si} is allowed to vary across firms but not over time.
- **Alternative approach.** The paper also uses a GMM specification in which productivity follows an AR(1) process and lagged instruments identify the CES parameters.
- **Estimated magnitudes.** Two-digit γ_s estimates range from 1.28 to 3.38 in the United States and from 1.38 to 2.05 in China. The weighted mean is 1.60 in the United States and 1.52 in China.
- **Calibrating σ.** The paper sets $\sigma = 1.77$ so that the standard deviation of the efficient size distribution in the United States matches the observed standard deviation. This treats the U.S. firm-size distribution as approximately efficient.

3.13 Recovering distortion-inclusive marginal debt and equity costs

From the firm's first-order conditions, the marginal cost of debt is recovered from

$$\frac{\alpha_s(\sigma-1)}{\sigma}\frac{P_{si}F_{si}}{\alpha_s D_{si}+(1-\alpha_s)D_{si}^{\frac{1}{\gamma_s}}E_{si}^{\frac{\gamma_s-1}{\gamma_s}}} = \frac{\partial}{\partial D_{si}}\left[(1+\tau^D_{si})r_{si}D_{si}+(1+\tau^E_{si})\lambda_{si}E_{si}\right], \tag{20}$$

and the marginal cost of equity from

$$\frac{(1-\alpha_s)(\sigma-1)}{\sigma}\frac{P_{si}F_{si}}{\alpha_s D_{si}^{\frac{\gamma_s-1}{\gamma_s}}E_{si}^{\frac{1}{\gamma_s}}+(1-\alpha_s)E_{si}} = \frac{\partial}{\partial E_{si}}\left[(1+\tau^D_{si})r_{si}D_{si}+(1+\tau^E_{si})\lambda_{si}E_{si}\right]. \tag{21}$$

- **Interpretation.** These are the "distortion-inclusive" marginal costs of debt and equity reported later in the empirical tables.
- **Why useful.** They allow the authors to study which firms face especially high financing distortions, even without separately identifying primitive wedges and primitive interest rates.

3.14 Labor-income extension

To allow liabilities to finance both capital and labor more explicitly, the paper studies the extension

$$F_{si} = A_{si}\left[\alpha_s D_{si}^{\frac{\gamma_s-1}{\gamma_s}}+(1-\alpha_s)E_{si}^{\frac{\gamma_s-1}{\gamma_s}}\right]^{\frac{\beta_s\gamma_s}{\gamma_s-1}} L_{si}^{1-\beta_s}, \tag{22}$$

with cost minimization

$$\min_{\{D_{si},E_{si},L_{si}\}} \left\{(1+\tau_{si}^D)r_{si}D_{si} + (1+\tau_{si}^E)\lambda_{si}E_{si} + (1+\tau_{si}^L)wL_{si}\right\}. \tag{23}$$

- **Interpretation.** This nests the possibility that liabilities fund labor as well as capital.
- **Key result.** The optimal debt–equity ratio is unchanged. Quantitative reallocation gains are also very similar to the baseline model.

3.15 Dynamic-model validation

- The paper also studies a Hennessy–Whited style dynamic model with tangible capital, intangible capital, labor, net debt, a tax benefit of debt, a collateral constraint, and an equity issuance cost.
- The main validation result is that the static misallocation measure rises monotonically with the equity issuance cost. So the paper's static CES framework does indeed detect stronger financial frictions in a standard dynamic corporate-finance environment.
- In the simulated model, the measure of misallocation tracks declines in market-to-book ratios and aggregate output reasonably well, and the estimated γ_s falls as frictions become more severe.
- The exercise also supports the interpretation that γ_s is largely shaped by sector-level technology and collateralizability rather than purely idiosyncratic firm shocks.

4 Identification and instruments (what makes the estimates credible)

4.1 This is a measurement framework, not a conventional causal design

- The paper does *not* identify the causal effect of one named policy or one named shock on financing. Instead, it measures how far the observed cross-section lies from the allocation implied by a structured benchmark model.
- Credibility therefore comes from the economic structure: the CES technology, monopolistic competition, the planner's problem, and the way these assumptions map observables into productivities and wedges.
- The gain estimate is only as good as the model's ability to approximate the true relation between liabilities and real output.

4.2 What identifies firm-specific distortions

- **Observed objects.** The data provide nominal value-added, total liabilities, and equity for each firm.
- **Model restriction.** Within a sector, firms share the same substitution parameters $(\sigma, \gamma_s, \alpha_s)$, so differences in the observed relation between financing and value-added are attributed to productivity A_{si} and wedges $(\tau_{si}^D, \tau_{si}^E)$.

- **Efficient benchmark.** At the planner's allocation, debt and equity are distributed in proportion to productivity, with a common sector leverage ratio. Deviations from that benchmark reveal distortions.

4.3 Why estimate γ_s separately from wedges

- Estimating γ_s separately uses variation in real value-added and financing choices to measure sector-wide substitutability between debt and equity.
- Once γ_s is fixed, the remaining within-sector dispersion is interpreted as firm-specific distortions rather than sector-wide average technological substitutability.
- This separation is important because otherwise "type" misallocation would mechanically reflect whatever value of γ_s best rationalizes the same leverage dispersion the wedges are supposed to explain.

4.4 Why normalize China relative to the United States

- Even in the United States, the model produces modest positive gains from reallocation. The authors interpret these gains as partly reflecting misspecification of a simple static framework.
- China-versus-U.S. normalization therefore removes some of the common-model error and produces a more credible estimate of the extra inefficiency in China.
- This is why the headline numbers in the abstract are the 51% to 69% gains relative to U.S. efficiency, rather than the raw China-only gains of roughly 69% to 89%.

4.5 Bootstrap inference

- The paper calculates standard errors both asymptotically and by bootstrap.
- The asymptotic standard errors are much smaller, so the reported bootstrap standard errors are deliberately conservative.
- Even then, the key gain estimates are precisely estimated because they are functions of sample means computed from very large firm-level samples.

4.6 What the dynamic validation adds

- The dynamic simulation exercise is not the main estimator, but it is a credibility check.
- It shows that the paper's static measure rises when an economically meaningful financing friction – the equity issuance cost – rises.
- It also shows that the static measurement framework closely approximates real output losses in that dynamic environment, which helps justify the paper's interpretation of "scale" misallocation as closely related to real factor misallocation.

5 Tables and figures

5.1 Table I: Chinese firm summary statistics

Read:

- Leverage rises strongly with firm size. The liabilities-to-assets ratio increases from roughly 0.48 in the smallest bin to about 0.57 in the largest bin.
- Smaller Chinese firms appear to generate much more value-added per unit of assets than larger Chinese firms. Descriptively, high-productivity firms do not seem to be the ones receiving the most finance.
- Over time, average leverage changes little, but value-added rises. This is consistent with a rapidly growing manufacturing sector operating under persistent financing distortions.

Table I

Chinese Firm Summary Statistics

Calculations are based on a sample of Chinese firms from the annual survey conducted by the National Bureau of Statistics of China (NBS) from 1999 to 2007. All firms are in the manufacturing sector and all have more than 5 million Chinese yuan (CNY) in sales. There are a total of 1,248,729 firm-year observations, and all variables are reported in millions of 2005 CNY. Panel A presents summary statistics by firm-size percentile. Panel B presents summary statistics by year.

Panel A: Summary Statistics by Size

Size percentile	Observations	Assets	Liabilities	Equity	Liabilities/ Assets	Value-added
0–5	62,497	1.9	1.0	1.0	0.483	1.6
5–15	124,849	3.8	2.1	1.8	0.536	2.0
15–30	187,328	6.4	3.5	2.9	0.549	2.5
30–50	249,697	11.2	6.2	5.0	0.555	3.5
50–70	249,750	21.6	11.9	9.6	0.552	5.4
70–85	187,304	46.3	25.5	20.8	0.551	9.7
85–95	124,870	120.3	66.6	53.7	0.553	22.2
95–100	62,434	859.3	493.1	365.8	0.569	138.8

Panel B: Summary Statistics by Year

Year	Observations	Assets	Liabilities	Equity	Liabilities/ Assets	Value-added
1999	51,646	75.6	42.3	33.4	0.566	11.3
2000	62,822	78.1	43.6	34.4	0.566	12.4
2001	78,893	73.1	39.9	33.1	0.554	11.8
2002	95,520	70.9	38.9	32.0	0.551	11.9
2003	119,292	72.5	40.8	31.7	0.548	12.6
2004	181,692	59.5	34.3	25.2	0.564	10.7
2005	190,022	68.8	39.3	29.4	0.545	12.9
2006	217,242	70.8	40.3	30.5	0.539	14.0
2007	251,600	71.9	40.9	31.0	0.535	15.7

Table II

U.S. Firm Summary Statistics

Calculations are based on a sample of manufacturing firms (SIC 2000 to 3999) from Compustat. The sample period is 1999 to 2007 and includes 14,158 firm-year observations. All variables are reported in millions of 2005 USD. Value-added is operating income before depreciation (OIBDP) plus imputed wages. Imputed wages are calculated by multiplying the employment of each firm with the mean wage per employee in the appropriate three-digit SIC industry. Panel A presents summary statistics by firm-size percentile. Panel B presents summary statistics by year.

Panel A: Summary Statistics by Size

Size percentile	Observations	Assets	Liabilities	Equity	Liabilities/ Assets	Value-added
0–5	713	9.9	3.3	6.6	0.330	4.7
5–15	1,416	25.6	9.7	15.9	0.378	11.0
15–30	2,123	63.5	22.9	40.6	0.369	25.2
30–50	2,829	170.1	62.7	107.4	0.367	62.1
50–70	2,833	507.8	226.8	281.0	0.438	188.0
70–85	2,124	1,480.5	796.7	683.8	0.540	518.0
85–95	1,416	4,707.0	2,802.9	1,904.1	0.591	1,543.1
95–100	704	28,128.4	16,154.0	11,974.5	0.596	8,169.6

Panel B: Summary Statistics by Year

Year	Observations	Assets	Liabilities	Equity	Liabilities/ Assets	Value-added
1999	1,936	1,479.1	883.7	595.4	0.466	507.6
2000	1,782	1,719.4	1,000.2	719.2	0.460	597.7
2001	1,612	2,013.2	1,172.5	840.7	0.450	614.3
2002	1,541	2,131.0	1,257.1	873.9	0.448	622.1
2003	1,544	2,263.7	1,286.4	977.3	0.430	663.4
2004	1,517	2,469.3	1,365.8	1,103.5	0.418	750.5
2005	1,470	2,578.9	1,411.5	1,167.4	0.424	811.2
2006	1,415	2,860.4	1,549.0	1,311.4	0.426	877.7
2007	1,341	3,112.7	1,695.0	1,417.7	0.431	912.2

5.2 Table II: U.S. firm summary statistics

Read:

- U.S. leverage also rises with firm size, but unlike China, the value-added-to-assets ratio looks much flatter across size bins.
- This is exactly the descriptive pattern one would expect in a relatively well-functioning financial system: larger firms are not obviously less productive users of resources.
- Over time, U.S. firms get larger and somewhat less leveraged, again contrasting with the Chinese sample.

5.3 Table III: Elasticity of substitution summary statistics

Read:

- The estimated debt–equity substitution elasticity is finite in both countries, with weighted means of about 1.60 in the United States and 1.52 in China.
- Whole-economy estimates from the Kmenta fixed-effects method and the alternative GMM method are very similar, which is reassuring.
- The main message is that the data do not look like a literal MM world with perfect substitutability between debt and equity.

Table III

Elasticity of Substitution Summary Statistics

Calculations are based on two samples of firms. One is a sample of U.S. firms from Compustat, and the other is a sample of Chinese firms from the National Bureau of Statistics of China. The sample period is 1999 to 2007. This table presents summary statistics from the two-digit sector-level estimation of the elasticity of substitution between debt and equity, γ_s. The last two rows show the whole economy γ estimate using the Kmenta (1967) method with fixed effects and using GMM, respectively.

	United States	China
Mininum two-digit γ_s	1.28	1.38
Maximum two-digit γ_s	3.38	2.05
Mean two-digit γ_s	1.72	1.52
Weighted mean two-digit γ_s	1.60	1.52
Whole economy γ (Kmenta)	1.57	1.50
Whole economy γ (GMM)	1.52	1.58

Table IV

Reallocation Gains by Year

Calculations are based on two samples of firms. One is a sample of U.S. firms from Compustat, and the other is a sample of Chinese firms from the National Bureau of Statistics of China. The sample period is 1999 to 2007. This table presents potential reallocation gains when the substitutability between debt and equity varies at the two-digit sector level, and the elasticity of substitution between the real benefit of firms in a sector is $\sigma = 1.77$. Column (1) presents the observed U.S. allocation of real value-added as a fraction of the optimal U.S. allocation, $F_{US}/\hat{F}_{US}$. Column (2) presents the corresponding percentage gain from moving from the observed to the optimal allocation. Columns (3) and (4) present analogous calculations for Chinese firms. Columns (5) and (6) show the Chinese efficiency ratio as a fraction of the U.S. efficiency ratio, $(F_{China}/\hat{F}_{China})/(F_{US}/\hat{F}_{US})$, and the corresponding percentage gains. Columns (7) and (8) provide a breakdown of total misallocation into the misallocation due to the scale of finance and due to the mix of debt and equity, holding scale fixed. Standard errors are in parentheses below each estimate.

	United States		China		United States vs. China			
Year	(1) Fractional Benefit	(2) Percent Gain	(3) Fractional Benefit	(4) Percent Gain	(5) Fractional Benefit	(6) Percent Gain	(7) Percent Scale	(8) Percent Type
1999	0.901	11.0	0.581	72.3	0.644	55.2	45.0	10.2
	(0.006)	(0.7)	(0.008)	(2.4)	(0.010)	(2.3)	(1.9)	(0.7)
2000	0.891	12.3	0.573	74.6	0.643	55.5	46.3	9.2
	(0.009)	(1.1)	(0.013)	(3.8)	(0.016)	(3.7)	(3.2)	(0.8)
2001	0.897	11.4	0.587	70.3	0.654	52.8	43.4	9.4
	(0.009)	(1.1)	(0.007)	(2.1)	(0.011)	(2.6)	(2.1)	(0.7)
2002	0.891	12.2	0.579	72.7	0.650	53.9	44.5	9.4
	(0.008)	(1.0)	(0.011)	(3.3)	(0.013)	(3.1)	(2.7)	(0.8)
2003	0.892	12.2	0.591	69.3	0.663	50.9	41.3	9.6
	(0.009)	(1.1)	(0.007)	(2.0)	(0.009)	(2.2)	(1.5)	(0.8)
2004	0.902	10.9	0.556	80.0	0.616	62.3	50.7	11.6
	(0.008)	(1.0)	(0.005)	(1.7)	(0.008)	(2.2)	(1.7)	(0.7)
2005	0.902	10.8	0.542	84.4	0.601	66.4	53.3	13.2
	(0.009)	(1.1)	(0.005)	(1.8)	(0.009)	(2.4)	(1.8)	(0.8)
2006	0.897	11.5	0.546	83.2	0.608	64.4	50.8	13.5
	(0.009)	(1.0)	(0.004)	(1.5)	(0.007)	(1.9)	(1.5)	(0.7)
2007	0.896	11.6	0.529	88.9	0.591	69.2	54.4	14.8
	(0.008)	(1.0)	(0.004)	(1.3)	(0.007)	(2.0)	(1.4)	(0.8)

5.4 Table IV: Baseline reallocation gains by year

Read:

- U.S. firms would gain only about 10.8% to 12.3% from moving to the efficient allocation. This is modest, not zero, and is part of the reason the paper uses the United States as a benchmark rather than assuming perfect efficiency.
- Chinese firms would gain roughly 69% to 89% in raw terms, and 51% to 69% relative to the U.S. benchmark.
- The most important decomposition is in columns (7) and (8): most of the gains come from the *scale* of finance, not from the debt–equity mix. The type component is only around 9% to 15%, depending on the year.
- This table is the paper's headline result. Financial deepening in China would matter mainly because it sends more resources to high-productivity firms, not mainly because it fine-tunes leverage ratios.

5.5 Figure 1: Efficient versus actual size densities

Read:

- In the United States, the observed and efficient size distributions are very similar. This is consistent with relatively small reallocation gains.
- In China, the efficient distribution has much fatter tails than the observed one. Many firms are too small relative to their productivity, while others are too large.
- The figure visually reinforces the central interpretation of Table IV: the problem is a distorted firm-size distribution tied to the distribution of finance.

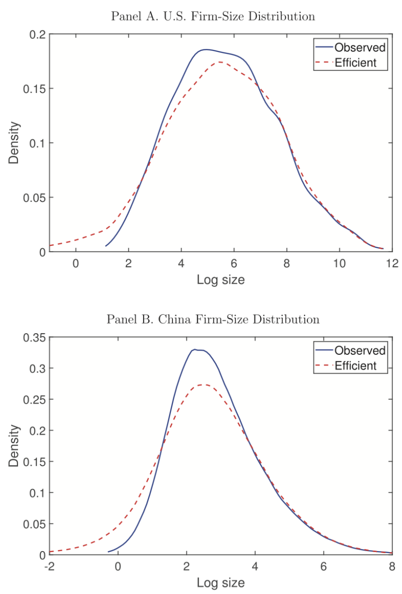

Figure 1. Efficient versus actual size densities. Panel A compares the U.S. observed and efficient firm-size distributions using a kernel density estimator. Observed firm size is computed as $\log(D_{si} + E_{si})$, and efficient firm size is computed as $\log(\hat{D}_{si} + \hat{E}_{si})$, where D_{si}, E_{si}, $\hat{D}_{si}$, and $\hat{E}_{si}$ are measured in millions of 2005 USD. Panel B similarly compares the observed and efficient firm-size distributions in China. Firm size is computed in the same manner, but D_{si}, E_{si}, $\hat{D}_{si}$, and $\hat{E}_{si}$ are measured in millions of 2005 CNY. (Color figure can be viewed at wileyonlinelibrary.com)

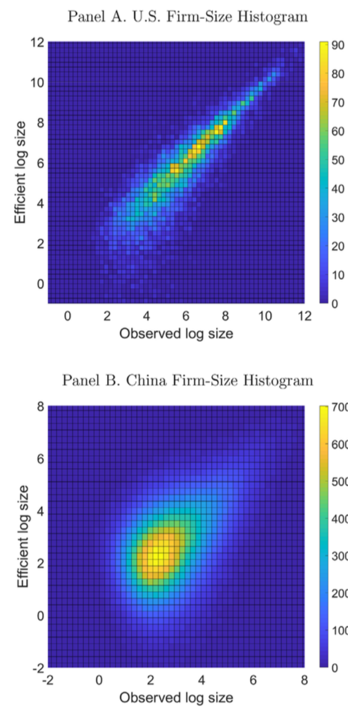

Figure 2. Efficient versus actual size histograms. Panel A contains the heat map of a three-dimensional histogram where the observed U.S. firm-size distribution is on the x-axis and the efficient U.S. firm-size distribution is on the y-axis. The legend for the z-axis heat map is located at the right of the plot and represents the number of observations in each bin. Observed firm size is computed as $\log(D_{si} + E_{si})$, and efficient firm size is computed as $\log(\hat{D}_{si} + \hat{E}_{si})$, where D_{si}, E_{si}, $\hat{D}_{si}$, and $\hat{E}_{si}$ are measured in millions of 2005 USD. Panel B similarly compares the observed and efficient firm-size distributions in China. Firm size is computed in the same manner, but D_{si}, E_{si}, $\hat{D}_{si}$, and $\hat{E}_{si}$ are measured in millions of 2005 CNY. (Color figure can be viewed at wileyonlinelibrary.com)

5.6 Figure 2: Efficient versus actual size histograms

Read:

- U.S. firms cluster near the 45-degree line, meaning actual and efficient sizes are close.
- Chinese firms are much more dispersed away from the 45-degree line, indicating substantial potential movement under efficient reallocation.
- The figure also suggests that smaller firms are more likely to be misallocated in both countries, though in China the distortions remain large across the whole size distribution.

5.7 Table V: Distortion-inclusive costs of debt and equity by year

Read:

- Mean debt and equity costs decline over time in the United States but rise in China.
- In China, the gap between means and medians is very large, especially later in the sample. This implies a strongly right-skewed cost distribution, with some firms effectively pushed far from financial markets.
- The time-series rise in Chinese financing costs lines up with the increase in measured misallocation after the 2004 NBS survey expansion.

Table V

Costs of Debt and Equity by Year

Calculations are based on two samples of firms. One is a sample of U.S. firms from Compustat, and the other is a sample of Chinese firms from the National Bureau of Statistics of China. The sample period is 1999 to 2007. This table displays the estimated mean (Panel A) and median (Panel B) distortion-inclusive marginal costs of debt and equity in the United States and China by year. All parameter settings are as in Table IV. Standard errors are in parentheses below each estimate.

	United States		China	
Year	Debt Cost	Equity Cost	Debt Cost	Equity Cost
Panel A: Mean Costs				
1999	0.232	0.189	0.191	0.164
	(0.0042)	(0.0117)	(0.0032)	(0.0039)
2000	0.227	0.185	0.205	0.167
	(0.0044)	(0.0043)	(0.0083)	(0.0030)
2001	0.214	0.172	0.203	0.173
	(0.0046)	(0.0045)	(0.0036)	(0.0034)
2002	0.217	0.176	0.202	0.179
	(0.0047)	(0.0062)	(0.0030)	(0.0027)
2003	0.220	0.173	0.270	0.173
	(0.0046)	(0.0053)	(0.0083)	(0.0014)
2004	0.222	0.169	0.226	0.182
	(0.0048)	(0.0039)	(0.0026)	(0.0017)
2005	0.218	0.171	0.288	0.202
	(0.0037)	(0.0040)	(0.0057)	(0.0022)
2006	0.213	0.167	0.314	0.204
	(0.0044)	(0.0044)	(0.0149)	(0.0012)
2007	0.206	0.166	0.355	0.228
	(0.0063)	(0.0060)	(0.0046)	(0.0017)
Panel B: Median Costs				
1999	0.188	0.154	0.085	0.096
	(0.0034)	(0.0022)	(0.0004)	(0.0005)
2000	0.184	0.153	0.091	0.102
	(0.0041)	(0.0024)	(0.0004)	(0.0004)
2001	0.168	0.137	0.095	0.106
	(0.0033)	(0.0029)	(0.0004)	(0.0004)
2002	0.171	0.138	0.098	0.107
	(0.0035)	(0.0029)	(0.0004)	(0.0004)
2003	0.171	0.134	0.102	0.109
	(0.0035)	(0.0030)	(0.0003)	(0.0004)
2004	0.177	0.138	0.101	0.112
	(0.0035)	(0.0027)	(0.0003)	(0.0003)
2005	0.177	0.141	0.112	0.117
	(0.0039)	(0.0027)	(0.0003)	(0.0002)
2006	0.169	0.139	0.117	0.122
	(0.0041)	(0.0025)	(0.0003)	(0.0003)
2007	0.162	0.130	0.127	0.131
	(0.0041)	(0.0030)	(0.0003)	(0.0003)

Table VI

Costs of Debt and Equity by Firm Size

Calculations are based on two samples of firms. One is a sample of U.S. firms from Compustat, and the other is a sample of Chinese firms from the National Bureau of Statistics of China. The sample period is 1999 to 2007. This table displays the estimated mean (Panel A) and median (Panel B) distortion-inclusive marginal costs of debt and equity in the United States and China by firm size. All parameter settings are as in Table IV. Standard errors are in parentheses below each estimate.

	United States		China	
Year	Debt Cost	Equity Cost	Debt Cost	Equity Cost
Panel A: Mean Costs				
0–5	0.355	0.202	0.832	0.468
	(0.0159)	(0.0127)	(0.0172)	(0.0045)
5–15	0.289	0.198	0.438	0.307
	(0.0090)	(0.0082)	(0.0068)	(0.0027)
15–30	0.249	0.181	0.334	0.240
	(0.0036)	(0.0043)	(0.0054)	(0.0014)
30–50	0.246	0.170	0.255	0.194
	(0.0027)	(0.0071)	(0.0034)	(0.0011)
50–70	0.214	0.166	0.210	0.158
	(0.0029)	(0.0025)	(0.0029)	(0.0017)
70–85	0.166	0.179	0.173	0.130
	(0.0018)	(0.0036)	(0.0030)	(0.0023)
85–95	0.146	0.169	0.131	0.105
	(0.0017)	(0.0032)	(0.0040)	(0.0011)
95–100	0.135	0.142	0.152	0.089
	(0.0021)	(0.0023)	(0.0557)	(0.0009)
Panel B: Median Costs				
0–5	0.266	0.151	0.338	0.286
	(0.0104)	(0.0063)	(0.0019)	(0.0011)
5–15	0.221	0.149	0.196	0.199
	(0.0050)	(0.0035)	(0.0007)	(0.0005)
15–30	0.207	0.140	0.143	0.156
	(0.0040)	(0.0026)	(0.0003)	(0.0003)
30–50	0.210	0.132	0.111	0.124
	(0.0031)	(0.0015)	(0.0003)	(0.0002)
50–70	0.176	0.138	0.090	0.099
	(0.0023)	(0.0020)	(0.0002)	(0.0002)
70–85	0.153	0.147	0.077	0.083
	(0.0019)	(0.0019)	(0.0002)	(0.0002)
85–95	0.136	0.151	0.068	0.074
	(0.0019)	(0.0024)	(0.0002)	(0.0002)
95–100	0.127	0.137	0.062	0.068
	(0.0039)	(0.0034)	(0.0003)	(0.0002)

5.8 Table VI: Distortion-inclusive costs by firm size

Read:

- In both countries, larger firms face lower financing costs.
- The size gradient is much steeper in China. For example, Chinese median debt costs fall dramatically from the smallest to the largest firms.
- This table gives the micro-foundation for the scale-misallocation result: productive but small firms appear to face especially high financial frictions in China.

5.9 Table VII: Firm characteristics and financing costs in China

Read:

- Firms in first-tier cities (Beijing, Shanghai, Shenzhen, Guangzhou) face significantly lower debt and equity costs.
- Larger firms face sharply lower financing costs, even after controlling for other observables.
- Conditional on size, firms with state investment actually have slightly *higher* costs on average. The paper notes that unconditional comparisons go the other way because state-connected firms are much larger.
- Foreign investment is associated with higher debt costs and essentially no meaningful reduction in equity costs. Young firms face slightly higher debt costs.

Table VII

Firm Characteristics and Debt and Equity Costs

Calculations are based on a sample of Chinese firms from the National Bureau of Statistics of China. The sample period is 1999 to 2007. This table presents two OLS regressions, in which the dependent variables are the distortion-inclusive marginal costs of debt and equity, respectively. The regressors are location, state investment, firm size, time, and firm age. *Location* is a dummy variable equal to one if a firm is located in Beijing, Shanghai, Shenzhen, or Guangzhou and zero otherwise. *State investment* is a dummy variable equal to one if a firm has a nonzero percentage of paid-in-capital from state sources and zero otherwise. *Foreign investment* is a dummy variable equal to one if a firm has a nonzero percentage of paid-in-capital from foreign sources and zero otherwise. *Size* is log total assets measured in 2005 CNY, *Time* is a linear time trend, and *Young* is a dummy variable equal to one if the firm is three or fewer years old and zero otherwise. t-statistics are in parentheses.

	Debt Cost	Equity Cost
Location	−0.054	−0.017
	(−5.8)	(−7.9)
State investment	0.019	0.011
	(0.9)	(2.2)
Foreign investment	0.088	−0.002
	(11.8)	(−1.3)
Size	−0.098	−0.057
	(−40.8)	(−104.6)
Time	0.022	0.007
	(16.5)	(24.7)
Young	0.015	−0.006
	(2.2)	(−3.7)

cities also face lower costs. Surprisingly, firms with nonzero state investment face slightly higher costs on average. It is important to note that this result is conditional on firm size. If we break down the full set of firms into those with and without state paid-in-capital, we find that firms with state paid-in-capital have lower costs. However, these firms are also significantly larger, so the effect of state investment on costs reverses after we control for firm size. Foreign investment, on the other hand, is associated with higher costs of debt but lower costs of equity, although this second effect is insignificant. We also find a positive coefficient on the time trend, which reflects the increasing costs also evident in Table V. Finally, we find that young firms face slightly highs of debt but slightly lower costs of equity.

Table VIII

Reallocation Gains by Elasticities of Substitution

Calculations are based on two samples of firms. One is a sample of U.S. firms from Compustat, and the other is a sample of Chinese firms from the National Bureau of Statistics of China. The sample period is 1999 to 2007. This table presents potential reallocation gains averaged across all years when we allow the elasticities of substitution, γ and σ, to vary. When γ is varied, σ is set to 1.77, and when σ is varied, γ_s is set to the benchmark two-digit sector values. Column (1) shows the observed U.S. allocation of real value-added as a fraction of the optimal U.S. allocation, $F_{\text{US}}/\hat{F}_{\text{US}}$. Column (2) shows the corresponding percentage gain from moving from the observed to the optimal allocation. Columns (3) and (4) present analogous calculations for Chinese firms. Columns (5) and (6) show the Chinese efficiency ratio as a fraction of the U.S. efficiency ratio, $(F_{\text{China}}/\hat{F}_{\text{China}})/(F_{\text{US}}/\hat{F}_{\text{US}})$, and the corresponding percentage gains. Columns (7) and (8) provide a breakdown of total misallocation into the misallocations due to the scale of finance and due to the mix of debt and equity, holding scale fixed. Standard errors are in parentheses below each estimate.

	United States		China		United States vs. China			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Parameter	Fractional Benefit	Percent Gain	Fractional Benefit	Percent Gain	Fractional Benefit	Percent Gain	Percent Scale	Percent Type
$\gamma = 1$	0.877	14.0	0.515	94.1	0.588	70.2	47.6	22.6
	(0.003)	(0.4)	(0.003)	(0.9)	(0.003)	(0.9)	(0.6)	(0.5)
γ_s	0.897	11.5	0.565	77.0	0.630	58.7	47.6	11.1
	(0.003)	(0.3)	(0.002)	(0.7)	(0.003)	(0.8)	(0.6)	(0.2)
$\gamma = 2$	0.903	10.8	0.583	71.6	0.646	54.9	47.6	7.2
	(0.002)	(0.3)	(0.002)	(0.7)	(0.003)	(0.7)	(0.6)	(0.2)
$\gamma = 5$	0.916	9.2	0.610	63.8	0.667	50.0	47.6	2.4
	(0.002)	(0.3)	(0.002)	(0.6)	(0.003)	(0.6)	(0.6)	(0.1)
$\gamma = 10$	0.920	8.7	0.618	61.7	0.672	48.7	47.6	1.1
	(0.002)	(0.3)	(0.002)	(0.6)	(0.003)	(0.6)	(0.6)	(0.0)
$\sigma = 1.5$	0.909	10.1	0.613	63.1	0.675	48.2	38.9	9.3
	(0.002)	(0.3)	(0.002)	(0.6)	(0.003)	(0.6)	(0.5)	(0.2)
$\sigma = 1.77$	0.897	11.5	0.565	77.0	0.630	58.7	47.6	11.1
	(0.003)	(0.3)	(0.002)	(0.7)	(0.003)	(0.8)	(0.6)	(0.2)
$\sigma = 2$	0.886	12.8	0.524	90.9	0.591	69.1	56.0	13.2
	(0.003)	(0.4)	(0.003)	(0.9)	(0.003)	(0.9)	(0.7)	(0.3)
$\sigma = 2.5$	0.862	16.0	0.435	129.8	0.505	98.1	78.3	19.8
	(0.004)	(0.5)	(0.003)	(1.6)	(0.004)	(1.5)	(1.1)	(0.5)
$\sigma = 3$	0.836	19.6	0.353	183.7	0.422	137.2	107.8	29.4
	(0.004)	(0.6)	(0.003)	(2.8)	(0.004)	(2.4)	(1.8)	(0.8)

5.10 Table VIII: Reallocation gains under alternative elasticities

Read:

- Changing γ has little effect on U.S. gains and some effect on Chinese gains, but the qualitative conclusion remains unchanged: scale dominates type.
- Changing σ matters a lot. As σ rises, reallocation gains become much larger because productive firms can expand more without being punished by steep demand curves.
- This is why the authors describe $\sigma = 1.77$ as conservative: it is at the low end of the plausible range.

5.11 Table IX: Reallocation gains for the U.S.–China size intersection

Read:

- Restricting the sample to the overlapping firm-size support cuts the relative Chinese gains roughly in half, to about 23% to 33%.
- But the surprising driver is not the removal of tiny Chinese firms. It is the removal of very large U.S. firms that are already close to efficient financing allocations.
- The interpretation is subtle: the lack of size overlap mostly masks misallocation among smaller U.S. firms rather than exaggerating misallocation in China.

Table IX

Reallocation Gains for the Firm-Size Intersection

Calculations are based on two samples of firms. One is a sample of U.S. firms from Compustat, and the other is a sample of Chinese firms from the National Bureau of Statistics of China. The sample period is 1999 to 2007. In each year, we limit the sample to the size intersection between U.S. and Chinese firms. Column (1) shows the observed U.S. allocation of real value-added as a fraction of the optimal U.S. allocation, $F_{\mathrm{US}}/\hat{F}_{\mathrm{US}}$. Column (2) shows the corresponding percentage gain from moving from the observed to the optimal allocation. The next two columns present analogous calculations for Chinese firms. The final two columns show the Chinese efficiency ratio as a fraction of the U.S. efficiency ratio, $(F_{\mathrm{China}}/\hat{F}_{\mathrm{China}})/(F_{\mathrm{US}}/\hat{F}_{\mathrm{US}})$, and the corresponding percentage gains. Standard errors are in parentheses below each estimate.

	United States		China		United States vs. China	
Year	(1) Fractional Benefit	(2) Percent Gain	(3) Fractional Benefit	(4) Percent Gain	(5) Relative Fractional Benefit	(6) Percent Gain
1999	0.817	22.4	0.616	62.3	0.754	32.6
	(0.011)	(1.5)	(0.010)	(2.7)	(0.016)	(2.7)
2000	0.792	26.2	0.643	55.4	0.812	23.2
	(0.014)	(2.1)	(0.007)	(1.6)	(0.016)	(2.4)
2001	0.798	25.4	0.639	56.4	0.801	24.8
	(0.011)	(1.6)	(0.007)	(1.7)	(0.013)	(2.1)
2002	0.797	25.5	0.629	59.1	0.789	26.8
	(0.012)	(1.8)	(0.012)	(3.0)	(0.019)	(3.1)
2003	0.792	26.3	0.638	56.8	0.805	24.2
	(0.012)	(1.8)	(0.006)	(1.5)	(0.015)	(2.3)
2004	0.808	23.8	0.613	63.2	0.759	31.8
	(0.012)	(1.8)	(0.006)	(1.5)	(0.014)	(2.5)
2005	0.817	22.4	0.625	59.9	0.765	30.7
	(0.013)	(1.8)	(0.006)	(1.5)	(0.014)	(2.4)
2006	0.825	21.3	0.628	59.3	0.761	31.3
	(0.011)	(1.6)	(0.005)	(1.3)	(0.012)	(2.1)
2007	0.819	22.1	0.616	62.2	0.753	32.9
	(0.012)	(1.8)	(0.005)	(1.4)	(0.012)	(2.2)

Table X

State-Owned Firms

Calculations are based on two samples of firms. One is a sample of U.S. firms from Compustat, and the other is a sample of Chinese firms from the National Bureau of Statistics of China. The sample period is 1999 to 2007. Column (1) shows the observed U.S. allocation of real value-added as a fraction of the optimal U.S. allocation, $F_{\mathrm{US}}/\hat{F}_{\mathrm{US}}$. Columns (2) to (4) report the Chinese state-owned firm efficiency ratio, $F_{\mathrm{China}}/\hat{F}_{\mathrm{China}}$, the Chinese state-owned firm efficiency ratio as a fraction of the U.S. efficiency ratio, $(F_{\mathrm{China}}/\hat{F}_{\mathrm{China}})/(F_{\mathrm{US}}/\hat{F}_{\mathrm{US}})$, and the corresponding percentage gains. Columns (5) to (7) repeat columns (2) to (4), except that calculations are based on a sample of all Chinese firms, private and state-owned. Standard errors are in parentheses below each estimate.

	United States	China State-Owned Firms			China All Firms		
Year	(1) Fractional Benefit	(2) Fractional Benefit	(3) Relative Fractional Benefit	(4) Percent Gain	(5) Fractional Benefit	(6) Relative Fractional Benefit	(7) Percent Gain
1999	0.901	0.614	0.681	46.8	0.587	0.652	53.5
	(0.006)	(0.010)	(0.012)	(2.7)	(0.007)	(0.009)	(2.0)
2000	0.891	0.611	0.686	45.8	0.577	0.647	54.5
	(0.009)	(0.013)	(0.016)	(3.5)	(0.010)	(0.013)	(3.2)
2001	0.897	0.587	0.654	52.8	0.569	0.634	57.8
	(0.009)	(0.015)	(0.019)	(4.3)	(0.010)	(0.013)	(3.3)
2002	0.891	0.635	0.712	40.4	0.590	0.662	51.1
	(0.008)	(0.011)	(0.014)	(2.7)	(0.008)	(0.011)	(2.5)
2003	0.892	0.654	0.733	36.4	0.602	0.675	48.1
	(0.009)	(0.011)	(0.014)	(2.5)	(0.007)	(0.010)	(2.2)
2004	0.902	0.649	0.720	39.0	0.574	0.637	57.0
	(0.008)	(0.014)	(0.017)	(3.2)	(0.005)	(0.008)	(2.0)
2005	0.902	0.640	0.709	41.0	0.556	0.616	62.3
	(0.009)	(0.018)	(0.020)	(4.0)	(0.007)	(0.009)	(2.5)
2006	0.897	0.654	0.729	37.1	0.561	0.626	59.8
	(0.009)	(0.017)	(0.019)	(3.5)	(0.005)	(0.008)	(2.0)
2007	0.896	0.651	0.727	37.6	0.543	0.607	64.8
	(0.008)	(0.018)	(0.022)	(4.1)	(0.005)	(0.008)	(2.3)

5.12 Table X: State-owned firms

Read:

- State-owned Chinese firms are still misallocated, but they show smaller percentage gains than the private Chinese sample.
- The paper interprets this as better market access for state-linked firms dominating their weaker profit-maximization incentives.
- When state-owned and private firms are pooled together, the percentage gains fall slightly, but the total gross amount of forgone output is still large because the measured economy is bigger.

Table XI
U.S. Reallocation Gains with Market Value Benefit

Calculations are based on a sample of U.S. firms from Compustat. The sample period is 1999 to 2007. The nominal benefit of finance is measured as the market value of debt plus the market value of equity, instead of value-added. This table presents potential reallocation gains when the substitutability between debt and equity varies at the two-digit sector level and the elasticity of substitution between the real benefit of firms in a sector is $\sigma = 1.77$. Column (1) shows the observed U.S. allocation of the real benefit of finance as a fraction of the optimal U.S. allocation, $F_{US}/\hat{F}_{US}$. Column (2) shows the corresponding percentage gain from moving from the observed to the optimal allocation. Standard errors are in parentheses below each estimate.

	United States	
Year	Fractional Benefit	Percent Gain
1999	0.728	37.4
	(0.026)	(4.8)
2000	0.739	35.4
	(0.024)	(4.2)
2001	0.853	17.3
	(0.012)	(1.6)
2002	0.879	13.8
	(0.011)	(1.4)
2003	0.878	13.9
	(0.012)	(1.5)
2004	0.877	14.1
	(0.012)	(1.6)
2005	0.874	14.4
	(0.013)	(1.7)
2006	0.885	13.0
	(0.009)	(1.1)
2007	0.867	15.4
	(0.011)	(1.4)

5.13 Table XI: U.S. gains using market value benefit instead of value-added

Read:

- Using market value of debt plus market value of equity as the nominal benefit of finance produces U.S. gains similar to the baseline in most years after 2001.
- The exception is the dot-com period, when estimated gains are much larger, around 35% to 37% in 1999–2000.
- The economic message is that easy access to public equity markets during the boom did not imply an efficient allocation of capital-market resources.

6 Short summary

- **Conclusion.** The paper extends the standard misallocation framework from real inputs to financial liabilities and finds that finance is substantially misallocated in China but only modestly misallocated in the United States.
- **Intuition.** The main distortion is not primarily the wrong leverage ratio. It is that productive firms do not receive enough total financing, so they operate at inefficiently small scale.
- **Empirical lesson.** When one compares China to a U.S. benchmark, the gains from fixing Chinese financial allocation are very large, but most of those gains come from sending more total funds to the right firms rather than from changing debt into equity or equity into debt.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Cross-sectional misallocation of financial liabilities is large in China and modest in the United States.
- **Headline magnitude.** Relative to the U.S. benchmark, China could raise real value-added by roughly 51% to 69% through within-sector reallocation of debt and equity.
- **Key decomposition.** About four-fifths of these gains come from the amount of finance, not the debt–equity mix.
- **Micro evidence on frictions.** Larger Chinese firms and firms in more developed cities face markedly lower distortion-inclusive financing costs.
- **Robustness.** The paper's qualitative message survives alternative elasticities, alternative samples, inclusion of state-owned firms, a labor-income extension, use of net debt, and a dynamic-model validation exercise.

7.2 Economic intuition

- In a well-functioning financial system, productive firms should be large because they should be able to attract the resources needed to operate at their efficient scale.
- A wrong debt–equity mix can matter, but only to the extent that debt and equity are imperfect substitutes. The paper finds that this channel is real but second-order relative to the overall quantity of funding.
- This is why the measured gains resemble the classic Hsieh–Klenow story: liability-side distortions map into real-side distortions because firms finance real inputs through their balance sheets.

7.3 Takeaway

This paper's central message is that the biggest financial-allocation problem in a developing economy need not be leverage composition per se. The more important issue can be that productive firms are starved of total financing while less productive firms are allowed to remain too large. By translating that idea into a tractable empirical framework, the paper gives a useful lower-bound measure of how much aggregate output depends on getting the cross-section of finance right.